



ANALYTICAL PROCEDURES

For DR/2000 and DR/3000 Instruments

PHOSPHORUS, Test 'N Tube™ Procedures

For water, wastewater, and seawater

Phosphorus, Total (0 to 3.50 mg/L PO₄³⁻; 0 to 1.10 mg/L P) Method 8190

Phosphorus, Reactive (0 to 5.00 mg/L PO₄³⁻; 0 to 1.60 mg/L P) Method 8048

Phosphorus, Acid Hydrolyzable (0 to 5.00 mg/L PO₄³⁻; 0 to 1.60 mg/L P) Method 8180

Method 8190

PHOSPHORUS, Total

For water, wastewater, and seawater

PhosVer® 3 Method and Acid Persulfate Digestion Test 'N Tube™ Procedure*

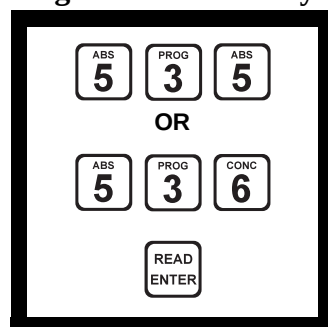
USEPA accepted for reporting wastewater analysis**



1. Turn on the COD Reactor. Heat to 150 °C. Place the plastic shield in front of the reactor.

Note: Ensure safety devices are in place to protect the analyst if splattering or leakage occurs.

Note: See COD Reactor manual for temperature adjustment instructions.



2. Enter the user-stored program number for reactive phosphorus, Test 'N Tube.

Press: **535 ENTER**

for mg/L PO₄³⁻ OR

Press: **536 ENTER**

for mg/L P

The display will show:

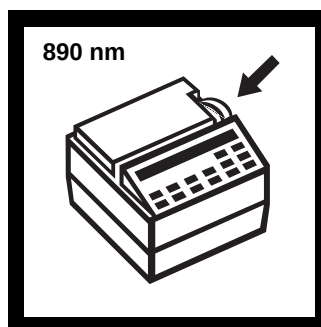
Dial nm to 890

Note: See Instrument Setup following these steps to enter this method into the DR/2000.

Note: DR/2000s with software versions 3.0 or greater will display **P** and the program number.

Note: Instruments with software versions 3.0 and greater will not display **DIAL nm TO** message if the wavelength is already set correctly. The display will show the message in step 4. Proceed with step 5.

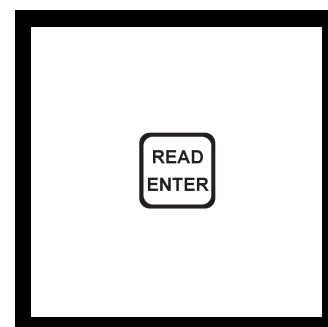
Note: For software versions 2.0 and 2.2, use Constant-on mode only.



3. Rotate the wavelength dial until the small display shows:

890 nm

Note: For DR/3000s, set wavelength to 890 nm and press **CLEAR**. Go to step 5.



4. Press: **READ/ENTER**

The display will show:

mg/L PO₄³⁻ Vial

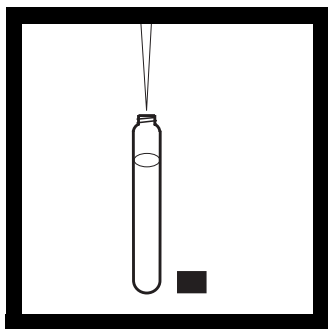
OR

mg/L P Vial

* Adapted from Standard Methods for the Examination of Water and Wastewater.

** Procedure is equivalent to USEPA Method 365.2 and Standard Method 4500-P E.

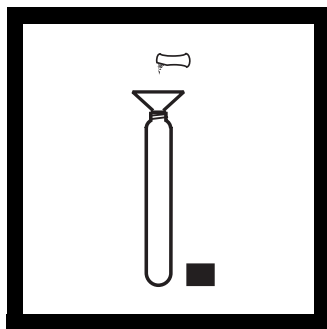
PHOSPHORUS, Total, continued



5. Use a TenSette® Pipet to add 5.0 mL of sample to a Total and Acid Hydrolyzable Test Vial.

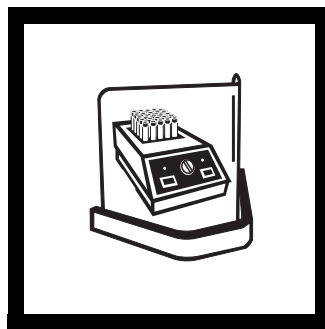
Note: For proof of accuracy, use a 1.0 mg/L Phosphate (0.33 mg/L P) Standard solution in place of the sample (see OPTIONAL REAGENTS).

Note: Run a reagent blank for this test. Use deionized water in place of the sample. Subtract this result from all test results from this method run with each lot of reagent.

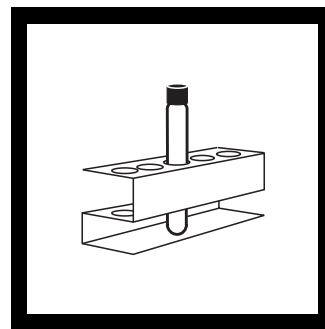


6. Using a funnel, add the contents of one Potassium Persulfate Powder Pillow for Phosphonate to the vial.

Cap tightly and shake to dissolve.

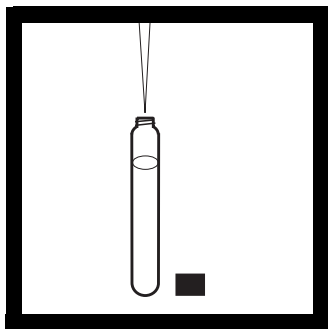


7. Heat the vial for 30 minutes at 150 °C.

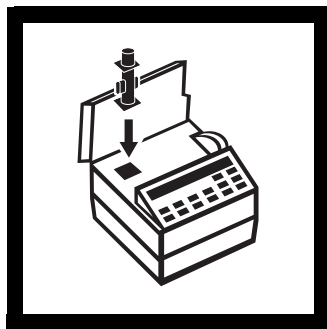


8. Carefully remove the vial from the reactor. Place it in a test tube rack and allow to cool to room temperature.

Note: Vials will be hot.

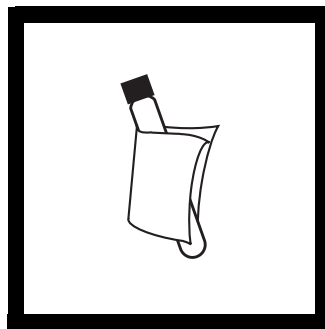


9. Use a TenSette Pipet to add 2.0 mL of 1.54 N sodium hydroxide to the vial. Cap and mix.



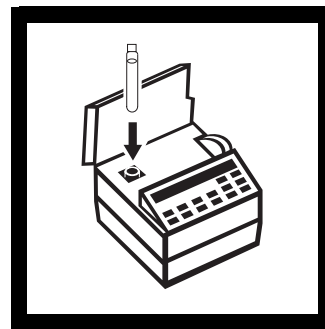
10. Place the COD Vial Adapter into the cell holder with the marker to the right.

Note: For the DR/3000, the groove in the adapter faces the front of the instrument.



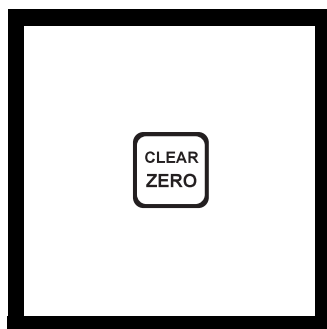
11. Clean the outside of the vial with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.



12. Place the sample vial in the adapter with the Hach logo facing the front of the instrument. Place the cover on the adapter.

PHOSPHORUS, Total, continued



13. Press: **ZERO**.

The display will show:

WAIT

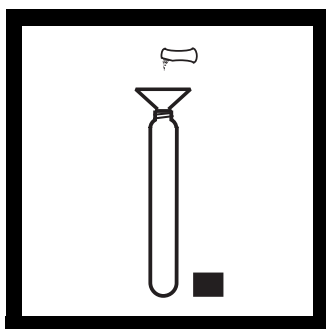
then: **0.00 mg/L PO₄³⁻ Vial**

OR

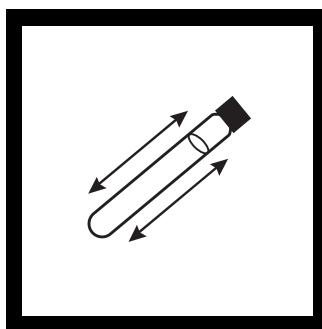
0.00 mg/L P Vial

Note: For DR/3000s, press **MANUAL PROGRAM**. Then press **ZERO** if the display does not show **0.000**. Enter the concentration factor (5.52 for mg/L PO₄³⁻ or 1.80 for mg/L P) and press **CONC FACTOR**. Then press 2 **CONC**.

Note: For multiple samples from the same source, zero only on the first sample. Read the remaining samples after adding the PhosVer 3 reagent. Subtract the PhosVer 3 reagent blank value from each reading.

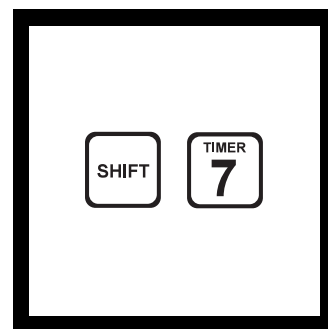


14. Using a funnel, add the contents of one PhosVer 3 Phosphate Reagent Powder Pillow to the vial.



15. Cap tightly and shake for 10-15 seconds.

Note: The powder will not completely dissolve.



16. Press: **SHIFT TIMER**

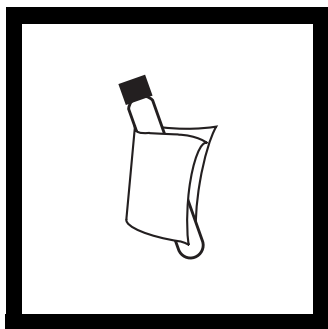
A 2-minute waiting period will begin.

Note: For DR/3000s, press 2 **TIMER**.

Note: Read samples between 2 and 8 minutes.

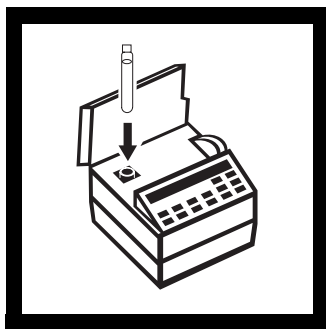
Note: DR/2000 instruments with a 10-minute color development time may be programmed with the reduced 2 minute development time. See Instrument Setup Section.

PHOSPHORUS, Total, continued

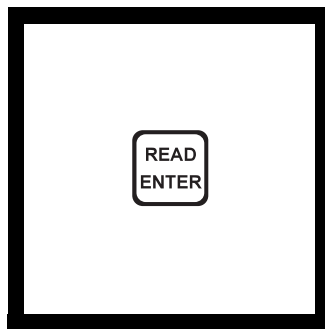


17. After the timer beeps, clean the outside of the sample vial with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.



18. Place the prepared sample vial into the adapter with the Hach logo facing the front of the instrument. Place the cover on the adapter.



19. Press: **READ/ENTER**

The display will show:

WAIT

then the display will show the results in mg/L PO_4^{3-} or mg/L P.

Note: For DR/3000s, press **CONC**.

Note: Results can be expressed as other forms of phosphorus:

$$\text{mg/L } \text{PO}_4^{3-} \times 3.07 = \text{mg/L P}$$

$$\text{mg/L } \text{PO}_4^{3-} \times 0.747 = \text{mg/L } \text{P}_2\text{O}_5$$

$$\text{mg/L P} \times 2.29 = \text{mg/L } \text{P}_2\text{O}_5$$

Note: In the **CONSTANT-ON** mode, pressing **READ/ENTER** is not required. **WAIT** will not appear. When the display stabilizes, read the result.

IMPORTANT NOTE: The test range for total phosphate is limited to 0 to 3.5 mg/L PO_4^{3-} . Values above 3.5 mg/L PO_4^{3-} may be used to estimate dilution ratios, but should **NOT** be used for reporting purposes.

PHOSPHORUS, Total, continued

REQUIRED REAGENTS

Total Phosphorus Test 'N Tube Reagent Set 50 tests..... 27426-45
Includes:

Description	Quantity Required		Unit	Cat. No.
	Per Test			
PhosVer 3 Phosphate Reagent Powder Pillows.....	1		50/pkg.....	21060-46
Potassium Persulfate Powder Pillows	1		50/pkg.....	20847-66
Sodium Hydroxide Solution, 1.54 N.....	2 mL		100 mL.....	27430-42
Total and Acid Hydrolyzable Test Vial	1		50/pkg.....	*
Water, deionized	100 mL		4 liters.....	272-56

REQUIRED APPARATUS

COD Reactor, 115/230 Vac (U.S.A. and Canada).....	1		each.....	45600-00
COD Reactor, 115/230 Vac (Europe)	1		each.....	45600-02
COD Vial Adapter, DR/2010	1		each.....	44799-00
Funnel, micro	1		each.....	25843-35
Pipet, TenSette, 1 to 10 mL	1		each.....	19700-10
Pipet Tips, for 19700-10 TenSette Pipet	2		50/pkg.....	21997-96
Pipet, TenSette, 0 to 1.0 mL	1		each.....	19700-01
Pipet Tips, for 19700-01 TenSette Pipet**	1		50/pkg.....	21856-96
Safety Shield, laboratory bench	1		each.....	23810-00
Test Tube Rack	1-3		each.....	18641-00

OPTIONAL REAGENTS

Total and Acid Hydrolyzable Test 'N Tube Reagent Set	50 tests.....	27427-45
Hydrochloric Acid Standard Solution, 6.0 N (1:1)	500 mL	884-49
Phosphate Standard Solution, 1 mg/L as PO_4^{3-}	500 mL	2569-49
Phosphate Standard Solution, Voluette Ampule, 50 mg/L as PO_4^{3-} , 10 mL	16/pkg.....	171-10
Sodium Hydroxide Standard Solution, 5.0 N.....	1 L.....	2450-53

OPTIONAL APPARATUS

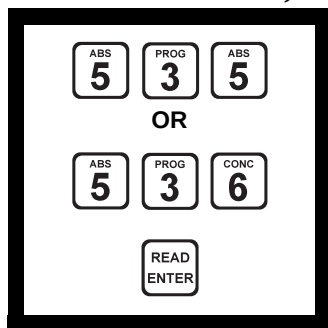
Cylinder, graduated, mixing, 25 mL	each.....	20886-40
pH Indicator Paper, 1 to 11 pH units	5 rolls/pkg.....	391-33
pH Meter, EC10, portable	each.....	50050-00

* These items are not sold separately.

** Larger sizes available.

PHOSPHORUS, Reactive

PhosVer® 3 Method; Test 'N Tube™ Procedure USEPA accepted for reporting wastewater analysis*



1. Enter the user-stored program number for reactive phosphorus, Test 'N Tube.

Press: **535 ENTER**

for mg/L PO_4^{3-}

OR

Press: **536 ENTER**

for mg/L P

The display will show:

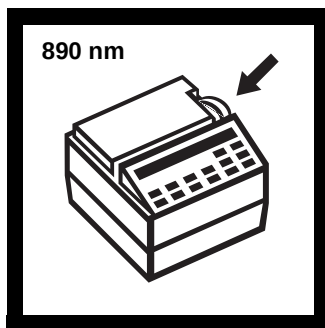
Dial nm to 890

Note: See Instrument Setup following these steps to enter this method into the DR/2000.

Note: DR/2000s with software versions 3.0 or greater will display **P** and the program number.

Note: Instruments with software versions 3.0 and greater will not display **DIAL nm TO** message if the wavelength is already set correctly. The display will show the message in step 4. Proceed with step 5.

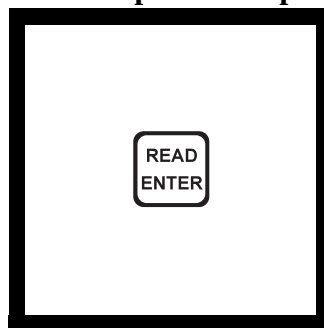
Note: For software versions 2.0 and 2.2, use Constant-on mode only.



2. Rotate the wavelength dial until the small display shows:

890 nm

Note: For DR/3000s, set wavelength to 890 nm and press **CLEAR**. Go to step 5.



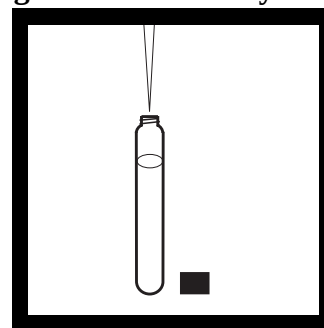
3. Press: **READ/ENTER**

The display will show:

mg/L PO_4^{3-} Vial

OR

mg/L P Vial



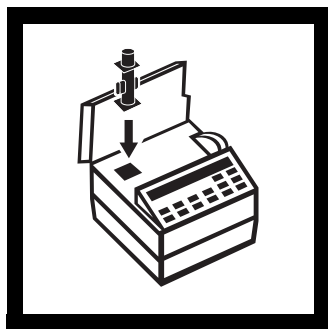
4. Use a TenSette® Pipet to add 5.0 mL of sample to a Reactive Phosphorus Test 'N Tube Dilution Vial. Cap and mix.

Note: For proof of accuracy, use a 1.0 mg/L Phosphate (0.33 mg/L P) Standard solution in place of the sample (see OPTIONAL REAGENTS).

Note: Run a reagent blank for this test. Use deionized water in place of the sample. Subtract this result from all test results from this method run with this lot of reagent.

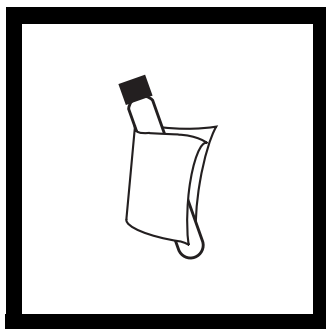
* Procedure is equivalent to USEPA Method 365.2 and Standard Method 4500-PB for wastewater.

PHOSPHORUS, Reactive, continued



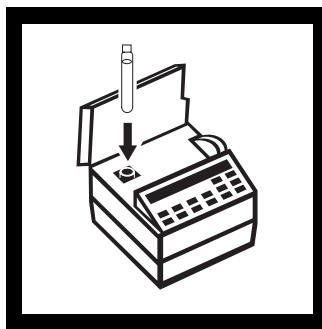
5. Place the COD Vial Adapter into the cell holder with the marker to the right.

Note: For the DR/3000, the groove in the adapter faces the front of the instrument.

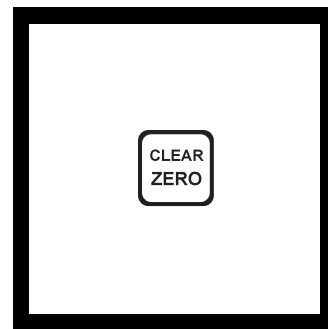


6. Clean the outside of the vial with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.



7. Place the vial into the adapter with the Hach logo facing the front of the instrument. Place the cover on the adapter.



8. Press: **ZERO**.

The display will show:

WAIT

then: **0.00 mg/L PO₄³⁻ Vial**

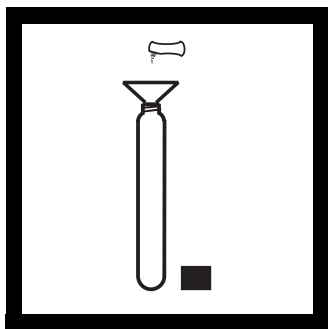
OR

0.00 mg/L P Vial

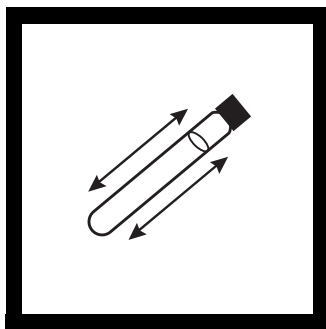
Note: For DR/3000s, press **MANUAL PROGRAM**. Then press **ZERO** if the display does not show **0.000**. Enter the concentration factor (5.52 for mg/L PO₄³⁻ or 1.80 for mg/L P) and press **CONC FACTOR**. Then press **2 CONC**.

Note: For multiple samples from the same source, zero only on the first sample. Read the remaining samples after adding the PhosVer 3 reagent. Subtract the PhosVer 3 reagent blank value from each reading.

PHOSPHORUS, Reactive, continued

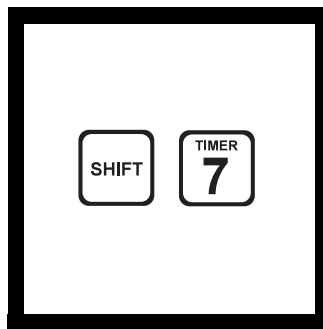


9. Using a funnel, add the contents of one PhosVer 3 Phosphate Reagent Powder Pillow to the vial.



10. Cap the vial tightly and shake for 10 to 15 seconds.

Note: The powder will not completely dissolve.



11. Press:

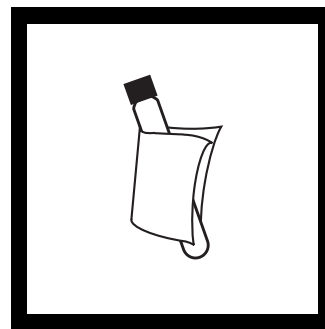
SHIFT TIMER

A 2-minute waiting period will begin.

Note: For DR/3000s, press **2 TIMER**.

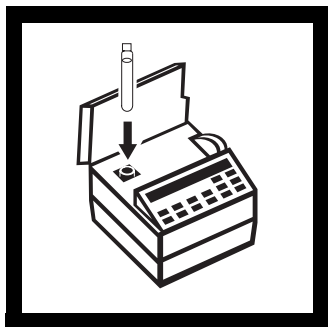
Note: Read samples between 2 and 8 minutes.

Note: DR/2000 instruments with a 10-minute color development time may be programmed with the reduced 2 minute development time. See Instrument Setup Section.

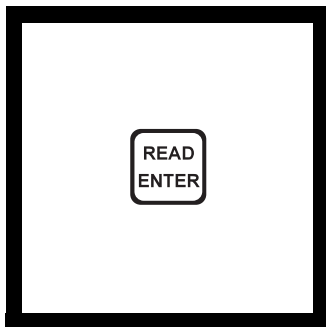


12. After the timer beeps, clean the outside of the sample vial with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.



13. Place the prepared sample vial into the adapter with the Hach logo facing the front of the instrument. Place the cover on the adapter.



14. Press: **READ/ENTER**

The display will show:

WAIT

then the display will show the results in mg/L PO₄³⁻ or mg/L P.

Note: For DR/3000s, press **CONC**.

Note: Results can be expressed as other forms of phosphorus:

$\text{mg/L PO}_4^{3-} \div 3.07 = \text{mg/L P}$
 $\text{mg/L PO}_4^{3-} \times 0.747 = \text{mg/L P}_2\text{O}_5$
 $\text{mg/L P} \times 2.29 = \text{mg/L P}_2\text{O}_5$

Note: In the CONSTANT-ON mode, pressing **READ/ENTER** is not required.

WAIT will not appear. When the display stabilizes, read the result.

PHOSPHORUS, Reactive, continued

REQUIRED REAGENTS

Reactive Phosphorus Test 'N Tube Reagent Set 50 tests..... 27425-45
Includes: (1) 21060-46, (50) Reactive Phosphorus Vials*

Description	Quantity Required		Unit	Cat. No.
	Per Test			
PhosVer 3 Phosphate Reagent Powder Pillows.....	1		50/pkg.....	21060-46
50 Reactive Phosphorus Test 'N Tube Dilution Vials.....	1		50/pkg.....	*

REQUIRED APPARATUS

DR/4000 Test Tube Adapter.....	1		each.....	48189-00
Funnel, micro	1		each.....	25843-35
Pipet, TenSette, 1 to 10 mL	1		each.....	19700-10
Pipet Tips, for 19700-10 TenSette Pipet	1		50/pkg.....	21997-96
Test Tube Rack.....	1-3		each.....	18641-00

OPTIONAL REAGENTS

Bromine Water, 30 g/L	29 mL**			2211-20
Hydrochloric Acid Standard Solution, 6.0 N (1:1)	500 mL.....			884-49
Phenol Solution, 30 g/L.....	29 mL.....			2112-20
Phosphate Standard Solution, 1 mg/L as PO_4^{3-}	500 mL.....			2569-49
Phosphate Standard Solution, PourRite Ampule, 50 mg/L as PO_4^{3-} , 10 mL.....	16/pkg.....			171-10
Phosphate Standard Solution, 50 mg/L	500 mL.....			171-49
Water, deionized	4 L.....			272-56

OPTIONAL APPARATUS

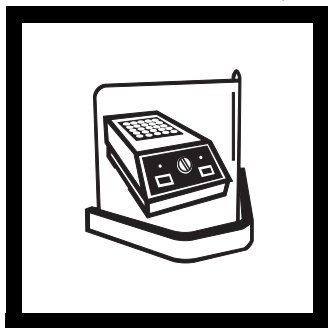
Flask, volumetric, Class A, 50-mL	each.....			14574-41
pH Paper, pH 1.0 to 11.0	5 rolls/pkg.....			391-33
pH Meter, EC10, portable	each.....			50050-00
Pipet, TenSette, 0.1 to 1.0 mL	each.....			19700-01
Pipet Tips, for 19700-01 TenSette Pipet	50/pkg.....			21856-96
Pipet Tips, for 19700-10 TenSette Pipet	1000/pkg.....			21997-28
Pipet, volumetric, Class A, 2.00-mL	each.....			14515-36
PourRite Ampule Breaker	each.....			24846-00

* These items are not sold separately.

** Larger sizes available.

PHOSPHORUS, Acid Hydrolyzable

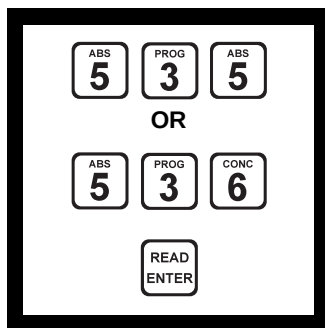
PhosVer® 3 Method; Test 'N Tube™ Procedure



1. Turn on the COD Reactor. Heat to 150 °C. Place the plastic shield in front of the reactor.

Note: Ensure safety devices are in place to protect the analyst if splattering or leakage occurs.

Note: See COD Reactor manual for temperature adjustment instructions.



2. Enter the user-stored program number for reactive phosphorus, Test 'N Tube.

Press: **535 ENTER**

for mg/L PO_4^{3-}

OR

Press: **536 ENTER**

for mg/L P

The display will show:

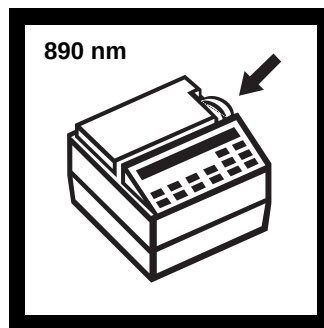
Dial nm to 890

Note: See Instrument Setup following these steps to enter this method into the DR/2000.

Note: DR/2000s with software versions 3.0 or greater will display **P** and the program number.

Note: Instruments with software versions 3.0 and greater will not display **DIAL nm TO** message if the wavelength is already set correctly. The display will show the message in step 4. Proceed with step 5.

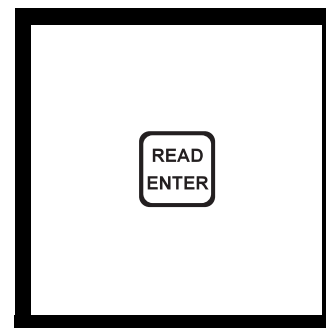
Note: For software versions 2.0 and 2.2, use Constant-on mode only.



3. Rotate the wavelength dial until the small display shows:

890 nm

Note: For DR/3000s, set wavelength to 890 nm and press **CLEAR**. Go to step 5.



4. Press: **READ/ENTER**

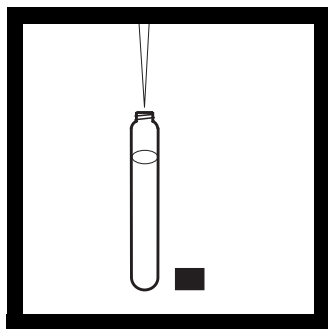
The display will show:

mg/L PO_4^{3-} Vial

OR

mg/L P Vial

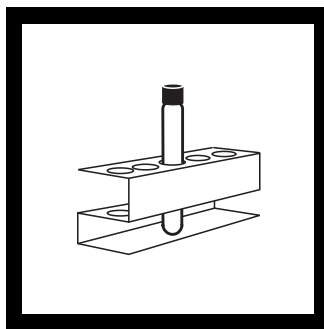
PHOSPHORUS, Acid Hydrolyzable, continued



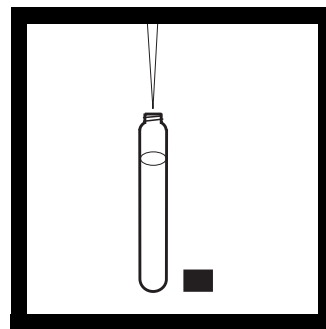
5. Use a TenSette® Pipet to add 5.0 mL of sample to a Total and Acid Hydrolyzable Test Vial. Cap and mix.



6. Heat the vial for 30 minutes at 150° C.



7. Carefully remove the vial from the reactor. Place it in a test tube rack and allow to cool to room temperature.

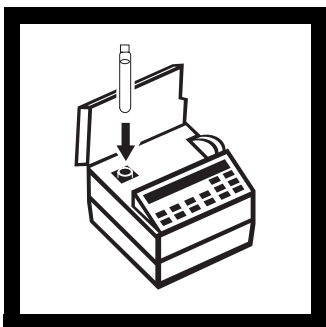


8. Use a TenSette Pipet to add 2.0 mL of 1.00 N sodium hydroxide to the vial. Cap and mix.

Note: For proof of accuracy, use a 1.0 mg/L Phosphate (0.33 mg/L P) Standard solution in place of the sample (see Optional Reagents).

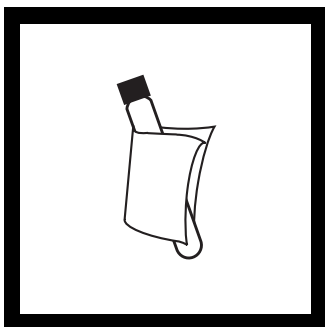
Note: Run a reagent blank for this test. Use deionized water in place of the sample. Subtract this result from all test results from this method run with this lot of reagent.

PHOSPHORUS, Acid Hydrolyzable, continued



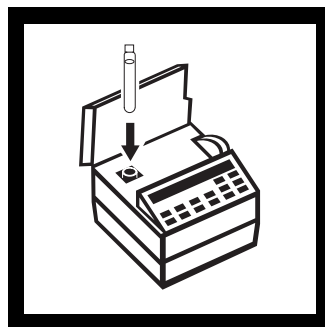
9. Place the COD Vial Adapter into the cell holder with the marker to the right.

Note: For the DR/3000, the groove in the adapter faces the front of the instrument.

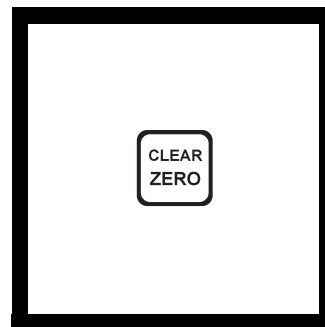


10. Clean the outside of the vial with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.



11. Place the sample vial in the adapter with the Hach logo facing the front of the instrument. Place the cover on the adapter.



12. Press: **ZERO**.

The display will show:

WAIT

then: **0.00 mg/L PO₄³⁻ Vial**

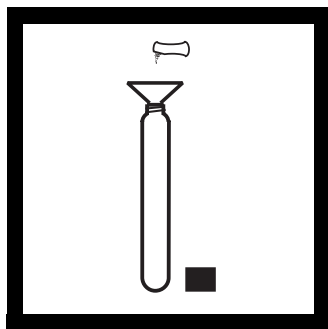
OR

0.00 mg/L P Vial

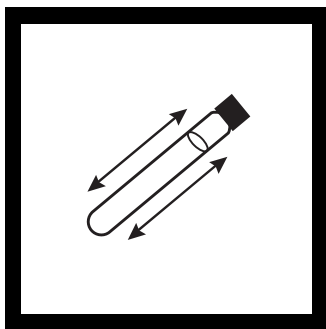
Note: For DR/3000s, press **MANUAL PROGRAM**. Then press **ZERO** if the display does not show **0.000**. Enter the concentration factor (5.52 for mg/L PO₄³⁻ or 1.80 for mg/L P) and press **CONC FACTOR**. Then press **2 CONC**.

Note: For multiple samples from the same source, zero only on the first sample. Read the remaining samples after adding the PhosVer 3 reagent. Subtract the PhosVer 3 reagent blank value from each reading.

PHOSPHORUS, Acid Hydrolyzable, continued

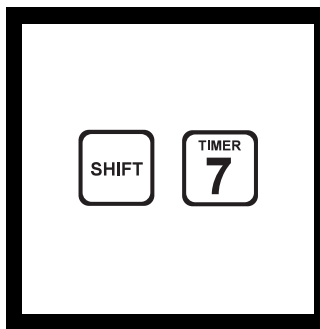


13. Using a funnel, add the contents of one PhosVer 3 Phosphate Reagent Powder pillow to the vial.



14. Cap tightly and shake for 10 to 15 seconds.

Note: The powder will not completely dissolve.



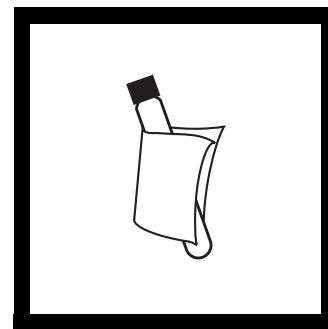
15. Press: **SHIFT TIMER**

A 2-minute waiting period will begin.

Note: For DR/3000s, press **2 TIMER**.

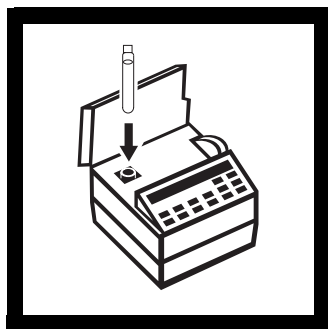
Note: Read samples between 2 and 8 minutes.

Note: DR/2000 instruments with a 10-minute color development time may be programmed with the reduced 2 minute development time. See Instrument Setup Section.

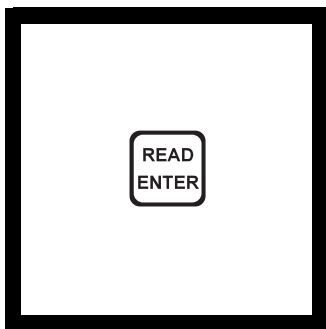


16. After the timer beeps, clean the outside of the sample vial with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.



17. Place the prepared sample vial into the adapter with the Hach logo facing the front of the instrument. Place the cover on the adapter.



18. Press: **READ/ENTER**

The display will show:

WAIT

then the display will show the results in mg/L PO_4^{3-} or mg/L P.

Note: For DR/3000s, press **CONC**.

Note: Results can be expressed as other forms of phosphorus:

$$\text{mg/L PO}_4^{3-} \div 3.07 = \text{mg/L P}$$

$$\text{mg/L PO}_4^{3-} \times 0.747 = \text{mg/L P}_2\text{O}_5$$

$$\text{mg/L P} \times 2.29 = \text{mg/L P}_2\text{O}_5$$

In the **CONSTANT-ON** mode, pressing **READ/ENTER** is not required.

WAIT will not appear. When the display stabilizes, read the result.

PHOSPHORUS, Acid Hydrolyzable, continued

REQUIRED REAGENTS

Total and Acid Hydrolyzable Test 'N Tube Reagent Set.....50 tests 27427-45
Includes: (1) 272-56, (1) 1045-42, (1) 20847-66, (1) 21060-46, (50) Total and Acid Hydrolyzable Test Vials*

Description	Quantity Required		Unit	Cat. No.
	Per Test			
PhosVer 3 Phosphate Reagent Powder Pillows	1	50/pkg	21060-46	
Potassium Persulfate Powder Pillows.....	1	50/pkg	20847-66	
Sodium Hydroxide Solution, 1.00 N	2 mL.....	100 mL	1045-42	
Water, deionized.....	100 mL.....	4 liters	272-56	
Total and Acid Hydrolyzable Test Vials	1	50/pkg	*	

REQUIRED APPARATUS

COD Reactor, 115/230 Vac (U.S.A. and Canada)	1	each	45600-00
COD Reactor, 115/230 Vac, 50 Hz (Europe).....	1	each	45600-02
DR/4000 Test Tube Adapter	1	each	48189-00
Funnel, micro	1	each	25843-35
Pipet, volumetric, Class A, 2.00-mL	1	each	14515-36
Pipet, volumetric, Class A, 5.00-mL	1	each	14515-37
Pipet Filler, safety bulb	1	each	14651-00
Pipet, TenSette, 1 to 10 mL.....	1	each	19700-10
Pipet Tips, for 19700-10 TenSette Pipet	1	1000/pkg	21997-28
Safety Shield, laboratory bench	1	each	23810-00
Test Tube Rack	1-3	each	18641-00

OPTIONAL REAGENTS

Bromine Water, 30 g/L.....	29 mL**	2211-20
Hydrochloric Acid Standard Solution, 6.0 N (1:1).....	500 mL	884-49
Phenol Solution, 30 g/L	29 mL	2112-20
Phosphate Standard Solution, 1 mg/L as PO_4^{3-}	500 mL	2569-49
Phosphate Standard Solution, 50 mg/L.....	500 mL	171-49
Phosphate Standard Solution, 2-mL PourRite Ampule, 50 mg/L as PO_4^{3-}	20/pkg	171-20
Sodium Hydroxide Standard Solution, 5.0 N	1 L	2450-53

OPTIONAL APPARATUS

pH Paper, pH 1.0 to 11.0.....	5 rolls/pkg	391-33
pH Meter, EC10, portable.....	each	50050-00
Pipet, TenSette, 0.1 to 1 mL.....	1.....each	19700-01
Pipet Tips, for 19700-01 TenSette Pipet.....	1.....50/pkg	21856-96
Pipet, volumetric, Class A, 2.00-mL	each	14515-36
Pipet, volumetric, Class A, 4.00-mL	each	14515-04
Pipet, volumetric, Class A, 6.00-mL	each	14515-06
Pipet, volumetric, Class A, 8.00-mL	each	14515-08
Pipet, volumetric, Class A, 10.00-mL	each	14515-38
PourRite Ampule Breaker.....	each	24846-00

* These items are not sold separately.

** Larger sizes available.

PHOSPHORUS Test 'N Tube Procedures

Sampling And Storage

Note: Correct results for volume additions (see Volume Additions, Section I of the DR/2000 Procedures Manual).

Collect samples in plastic or glass bottles that have been acid cleaned with 1:1 Hydrochloric Acid Solution and rinsed with deionized water. Do not use commercial detergents containing phosphate for cleaning glassware used in this test.

Analyze samples immediately after collection for best results. If prompt analysis is impossible, use the preservation technique appropriate to the parameter you are measuring.

1. Total Phosphorus—Preserve for up to 28 days by adjusting the pH to 2 or less with concentrated sulfuric acid (about 2 mL per liter) and refrigerating at 4 °C. Neutralize with sodium hydroxide and warm the sample to room temperature before analysis.
2. Reactive Phosphorus—Preserve samples for up to 48 hours by filtering immediately and storing at 4 °C. Warm samples to room temperature before analysis.
3. Acid Hydrolyzable Phosphorus—Preserve samples for up to 24 hours by storing at 4 °C. Warm samples to room temperature before analysis.

Accuracy Check

Note: Clean glassware with 1:1 hydrochloric acid solution. Rinse again with deionized water. Do not use detergents containing phosphates to clean glassware.

Standard Additions Method

- a. Fill three 25-mL graduated mixing cylinders with 25 mL of sample.
- b. Snap the neck off a Phosphate Voluette Ampule Standard, 50 mg/L as PO_4^{3-} (Cat. No. 171-10).
- c. Use a TenSette Pipet to add 0.1 mL, 0.2 mL and 0.3 mL, respectively, to the three 25-mL aliquots of a water sample prepared in step a. Mix well.
- d. Analyze samples from step c as described in the procedure. Use 5.0 mL of the prepared sample for each test. The concentration should increase: 0.2 mg/L, 0.4 mg/L, 0.6 mg/L PO_4^{3-} , respectively.
- e. If these increases do not occur, see Section I of the DR/2000 Procedures Manual for more information.

Standard Solution Method

To check accuracy, use a 1.0 mg/L Phosphate Standard Solution listed under Optional Reagents. Or, prepare by pipetting 2.0 mL of solution from a Voluette Ampule Standard for Phosphate, 50 mg/L as PO_4^{3-} , into a Class A 100-mL volumetric flask. Dilute to the mark with deionized water. Substitute this standard for the sample and perform the procedure as described. The mg/L PO_4^{3-} reading should be 1.0 mg/L.

PHOSPHORUS Test 'N Tube Procedures, continued

Interferences

Large amounts of sample turbidity may cause inconsistent results in the test because the acid present in the powder pillows may dissolve some of the suspended particles and because of variable desorption of orthophosphate from the particles.

The PhosVer 3 Phosphate Reagent Powder Pillows should be stored in a cool, dry environment.

The following may interfere when present in concentrations exceeding these listed below:

Table 1

Substance	Interference Level
Aluminum	200 mg/L
Arsenate	Interferes at any level.
Chromium	100 mg/L
Copper	10 mg/L
Iron	100 mg/L
Nickel	300 mg/L
Silica	50 mg/L
Silicate	10 mg/L
Sulfide	6 mg/L*, 9 mg/L**, 90 mg/L***
Turbidity	May cause inconsistent results because the acid present in the powder pillows may dissolve some of the suspended particles and because of variable desorption of orthophosphate from the particles.
Zinc	80 mg/L
Highly buffered samples or extreme sample pH	May exceed the buffering capacity of the reagents and require sample pretreatment. See <i>Section I, pH Interferences</i> in the <i>DR/2000 Procedures Manual</i> for more information.

* Reactive Phosphorus

** Acid Hydrolyzable Phosphorus

*** Total Phosphorus

PHOSPHORUS Test 'N Tube Procedures, continued

Method Performance

Precision: Total Phosphorus

Standard: 3.00 mg/L PO_4^{3-}

Program	95% Confidence Limits
535	2.94 - 3.06 mg/L PO_4^{3-}

Precision: Reactive Phosphorus

Standard: 5.00 mg/L PO_4^{3-}

Program	95% Confidence Limits
535	4.92 - 5.08 mg/L PO_4^{3-}

Precision: Acid Hydrolyzable Phosphorus

Standard: 3.00 mg/L PO_4^{3-}

Program	95% Confidence Limits
535	2.96 - 3.04 mg/L PO_4^{3-}

Estimated Detection Limit

The Estimated Detection Limit (EDL) for this test is 0.04 mg/L PO_4^{3-} . The EDL is the calculated lowest average concentration in a deionized water matrix that is different from zero with a 99% level of confidence. For more information on derivation and use of Hach's estimated detection limit, see *Section I* of the *DR/2000 Procedures Manual*.

Sample Disposal Information

Final samples will contain molybdenum. In addition, final samples will have a pH less than 2 and are considered corrosive (D002) by the Federal RCRA. Consult the Material Data Safety Data Sheet for information specific to the reagents used.

Summary of Method

Phosphates present in organic and condensed inorganic forms (meta-, pyro- or other polyphosphates) must be converted to reactive orthophosphate before analysis. Pretreatment of the sample with acid and heat provides the conditions for hydrolysis of the condensed inorganic forms. Organic phosphates are converted to orthophosphates by heating with acid and persulfate.

Orthophosphate reacts with molybdate in an acid medium to produce a phosphomolybdate complex. Ascorbic acid then reduces the complex, giving an intense molybdenum blue color.

INSTRUMENT SETUP

DR/2000 with 1.261 and 1.27 software

Enter the calibration as an operator-programmed calibration. Follow the steps in the Operation section of the DR/2000 Instrument Manual.

Store the method as follows:

nm = 890

Decimal = 00.00

Units = mg/l

Symbol = P Vial OR PO_4^{3-} Vial (select PO_4^{3-} for phosphate or P for phosphorus)

Timer 1 = 2:00

Enter the calibration with 0.000 absorbance values for zero and #1 standard. To do this, do not place anything in the sample cell compartment. Begin by storing standards 0 and 1 as the concentrations shown in the table below. Accept 0.000 Abs as the absorbance value for all standards. Store the calibration values by pressing **SHIFT READ/ENTER**.

Next, edit the absorbance values for the standards to the values given below. Follow the steps given in the Operation section of the *DR/2000 Instrument Manual*.

Method	Standard	Concentration	Absorbance
535 (PO_4^{3-})	0	0.00	0.000
	1	5.50	0.962
Method	Standard	Concentration	Absorbance
536 (P)	0	0.00	0.000
	1	1.80	0.972

The method is now stored as an operator-programmed method number between 950 and 999. Record the method number for future reference when using this method.

INSTRUMENT SETUP, continued

DR/2000 with Software Versions 2.0 and 2.2

Enter the calibration as an update to a Hach stored program.

1. Press: **I/O**
2. Press: **SHIFT CONFIG/METH**
3. Press: **535 READ/ENTER** for mg/L PO₄³⁻ OR **536 READ/ENTER** for mg/L P.
4. Within 3 seconds, press: **SHIFT 3 CONFIG/METH**.
The display will show: **ENTER nm**
5. Press: **890 READ/ENTER**

If you make an error, press **SHIFT CLEAR** and re-enter the number.
When the number is correct, press **READ/ENTER**.
The display will show: **DECIMAL? 00.00**
6. The decimal point is correctly positioned. Press **READ/ENTER**.
7. The display will show: **UNITS?**
8. Use the arrow keys to select the appropriate unit of measure. Press the down arrow key once. The display will show: **mg/L**
9. Press: **READ/ENTER** when the correct unit of measure is displayed.
The display will show: **SYMBOL?**
10. Construct the display to read the correct symbol (either PO₄³⁻ Vial or P Vial):
 - a. Select letters and regular numbers by scrolling to the correct symbol with the arrow keys.
 - b. To make a letter uppercase, press the **SHIFT** key.
 - c. To enter subscript numbers, press the digit on the numeric keypad.
 - d. To enter superscript numbers, press the digit on the numeric keypad, then make it superscript by pressing **SHIFT**.
 - e. The space is the character displayed after one press of the **DOWN ARROW**.
 - f. Accept each symbol by pressing **READ/ENTER**.
 - g. To end symbol entry, press **READ/ENTER** a second time after accepting the last character.
11. When the instrument is out of symbol entry mode,
the display will show: **TIMER?**
12. This method has one timed step, so press **SHIFT TIMER**.
The display will show: **MM:SS TIME 1?**
13. To enter the timer value of 2 minutes, press: **0200**.
14. Press **READ/ENTER** to accept the timer value.
The display will show: **MM:SS TIME 2?**

INSTRUMENT SETUP, continued

15. Press **READ/ENTER** to complete timer entry mode.

The display will show: **#1 Data**

16. Enter the following 12 numbers as shown. Complete each number entry by pressing the **READ/ENTER** key.

	For mg/L PO ₄ ³⁻ (#535)	For mg/L P (#536)
# 1 Data	0	0
# 2 Data	9251	3083
# 3 Data	9252	3083
# 4 Data	9251	3083
# 5 Data	9252	3084
# 6 Data	9251	2828
# 7 Data	9252	2828
# 8 Data	9251	2828
# 9 Data	9252	3083
# 10 Data	5461	16384
# 11 Data	544	544
Checksum	51055	24708

The final number is a check value which determines if the data sequence was correctly entered. If an error was made during number entry, the display will return to the prompt for data # 1 and the entire sequence must be re-entered. If all the numbers are correctly entered, the display will return to the method prompt and is ready for use.

INSTRUMENT SETUP, continued

Note: If you have previously installed method 535 or 536 with a 10-minute timer interval, you must delete that method from memory before proceeding with this installation. To delete the previous method 535 or 536:

DR/2000 with Software Version 3.0 and 3.1

1. Turn the instrument on. Press: **SHIFT METHOD** to enter configuration mode. The display will show: **MOMENTARY** or **CONSTANT ON**
2. Press the up arrow key twice to select **HACH UPDATE**. Press **READ/ENTER**. The display will show: **ENTER #:**
3. Press the down arrow key once. The display will show: **ERASE #:**

Press **535** (for mg/L PO₄³⁻) or **536** (for mg/L P). then press **READ/ENTER**. The display will show: **P535:** for mg/L PO₄³⁻ OR **P536:** for mg/L P
4. Press **4200** (for mg/L PO₄³⁻) OR **3702** (for mg/L P).
5. Press **READ/ENTER**. the display will show: **COMPLETED**

The old installation of method 535 or 536 has been deleted. You can now install the updated procedure.

1. Turn the instrument on. Press: **SHIFT METHOD** to enter configuration mode. The display will show: **MOMENTARY** or **CONSTANT ON**
2. Press the up arrow key twice to select **HACH UPDATE**. Press **READ/ENTER**. The display will show: **ENTER #:**
3. Press **535** (for mg/L PO₄³⁻) or **536** (for mg/L P). Then press **READ/ENTER**. The display will show: **P535 ENTER nm** OR **P536 ENTER nm**
4. Press **890 READ/ENTER** (for mg/L PO₄³⁻) OR **3702** (for mg/L P)

The display will show: **P535 DECIMAL? 00.00** OR **P536 DECIMAL? 00.00**
5. The decimal point is already positioned correctly, so press **READ/ENTER**. The display will show: **P535 UNITS? OR P536 UNITS?**
6. Use the arrow keys to select the appropriate unit of measure. For this method, press the down arrow key twice.

The display will show: **P535 mg/L** OR **P536 mg/L**

7. With the correct unit of measure displayed, press **READ/ENTER**. The display will show: **P535 mg/L** OR **P536 mg/L**
8. Construct the display to read the correct symbol (either PO₄³⁻ Vial for method #535 or P Vial for method #536). The symbol must be entered exactly as shown, including dashes and spaces between characters.
 - a. Select letters and regular numbers by scrolling to the correct symbol with the arrow keys.
 - b. To make a letter upper case, press the **SHIFT** key.
 - c. To enter subscript or superscript numbers, press the appropriate number on the numeric keypad, then press **SHIFT** until the symbol is correct.

Note: If you make an error press **SHIFT CLEAR** and re-enter the number. When the number is correct, press **READ/ENTER**.

INSTRUMENT SETUP, continued

- d. Make sure you enter the line exactly as shown, including dashes. Do not enter trailing spaces.
 - e. To enter a space, press the down arrow key once.
 - f. Accept each symbol by pressing **READ/ENTER**.
 - g. To end symbol entry, press **READ/ENTER** a second time after the last character has been accepted.
9. When the instrument is out of symbol entry mode, the display will show:
P535 TIMER OR P536 TIMER
10. The method has one timed step. Press **SHIFT TIMER**. The display will show:
P535 MM:SS TIME 1? OR P536 MM:SS TIME 1?
11. To enter the timer value of 2 minutes, press **0200**.
12. Press **READ/ENTER** to accept the timer value. The display will show:
P535 MM:SS TIME 2? OR P536 MM:SS TIME 2?
13. Press **READ/ENTER** to complete timer entry.
The display will show: **# 0 STANDARD**
14. Press **READ/ENTER** to display the zero data pair.
The display will show: **0.000 Abs 00.00 mg/L**
15. Press **READ/ENTER**. The display will show: **# 1 STANDARD**
16. Press **READ/ENTER**. The display will prompt for entry of the first concentration point: **# 1 00.00 mg/L**
17. Enter concentration point #1 from the table below by pressing **0550** (for #535) or **0180** (for #536) so the display shows: **# 1 05.50 mg/L OR # 1 01.80 mg/L**
18. Press **READ/ENTER**. The display will prompt for entry of the first absorbance point: **# 1 0.000 Abs**
19. Enter the absorbance point from the table below by pressing **0962** for #535 or **0972** for #536 so the display shows: **# 1 0.962 Abs OR # 1 0.972 Abs**
20. Press **READ/ ENTER**. The display will show the first data pair: **0.962 Abs 05.50 mg/L** (for method 535) OR **0.972 Abs 01.80 mg/L** (for method 536)
21. Press **READ/ENTER** to accept the first data pair.
The display will show: **# 2 STANDARD**

INSTRUMENT SETUP, continued

The data pair values from the table below are now entered.

For Method 535

Standard	Concentration	Absorbance
# 0	0.00 [mg/l]	0.000 [Abs]
# 1	5.50 [mg/l]	0.962 [Abs]

For Method 536

Standard	Concentration	Absorbance
# 0	0.00 [mg/l]	0.000 [Abs]
# 1	1.80 [mg/l]	0.972 [Abs]

22. Press **SHIFT READ/ENTER** to complete data entry. The display will show: **#:**

23. Enter the validation number **3707** for method #535 or **3209** for method #536 so the display shows: **#: 3707 OR #: 3209**

24. Press **READ/ENTER**. The display will show: **COMPLETED** then:
P535 mg/L PO₄³⁻ Vial OR P536 mg/L P-Vial

Note: If the display shows: **INCORRECT #** then prompts again for the validation number, you may have made an error during data entry. Make sure the validation number is correct. If so, then the error occurred during another portion of the method entry. Press **METH** and respond to the **ABORT?** message by pressing **READ/ENTER**. Then re-enter the method.

The instrument is now ready for use with method 535 or 536.



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